# RITIK POKHAREL

GIS Analyst | Enterprise GIS | UAV Remote Sensing & ML
Baton Rouge, LA | 225-220-2592 | Email | LinkedIn | GitHub | Extended CV

## PROFESSIONAL SUMMARY

GIS Analyst with an M.S. in GIS, remote sensing, and data science delivering production Esri solutions across enterprise administration, offline field operations, web applications, spatial analysis, and technical training. Combines ArcGIS Pro, Enterprise, Python, UAV photogrammetry, and machine learning to turn complex spatial data into reliable client and field workflows. Authored 3 peer-reviewed papers in Q1 journals (Smart Agricultural Technology and Remote Sensing), communicating results to stakeholders and GIS professionals.

## TECHNICAL SKILLS

**GIS & Esri:** ArcGIS Pro, ArcGIS Enterprise (Portal, Server), ArcGIS Online, Experience Builder, Dashboards, Survey123, Field Maps, ModelBuilder, Spatial Analyst, Image Analyst, QGIS

**Programming & Data:** R, Python (ArcPy, API for Python, pandas, NumPy, GeoPandas, scikit-learn), SQL, GitHub, Jupyter, Arcade, JavaScript, QA/QC, EDA

**Remote Sensing & GeoAI:** UAV multispectral acquisition, Pix4D Mapper, ArcGIS Drone2Map, orthomosaics, DSM/DTM, radiometric calibration, GCPs, NDVI/NDRE, Sentinel-2, classification, change detection

**Machine Learning:** Random Forest, XGBoost, gradient boosting, neural networks (TensorFlow/Keras), SVM, feature engineering, cross-validation, independent-season validation

**Field & Cartography:** GNSS/GPS data collection, offline mobile deployment, map series, coordinate systems and projections, topology, symbology, print-ready layouts

## PROFESSIONAL EXPERIENCE

**GIS Analyst - Global Geospatial Solutions, LLC / Global Geospatial Institute | Jun 2026 - Present**

- Administer ArcGIS Enterprise, a geospatial data server, and an ArcGIS Online organization (1,400+ users, 500+ groups) for clients: publishing hosted layers, managing access/sharing, managed hosting, and automated bulk provisioning via the ArcGIS API for Python.
- Built a parish-scale offline ArcGIS Field Maps disaster-response system for Terrebonne Parish: 16 downloadable areas, ~10 operational layers, and 9-12 quad grid cells per area for fully disconnected field editing.
- Developed the Gardere Community Hub in Experience Builder and Survey123 (4 operational layers, 60+ community resources); presented the deployed app to stakeholders in Oct 2026.
- Delivered hands-on ArcGIS Pro, ArcGIS Online, Survey123, and Field Maps training to 300+ students, 15 teachers, and 19 industry professionals; built reusable exercise datasets and lab guides for K-12 and workforce programs.
- Ran a GIS Maturity Assessment across 50 weighted criteria with an executive summary and prioritized roadmap; produced client-ready maps, map books, dashboards, and stakeholder presentations.

**Graduate Research Assistant - Louisiana State University | May 2024 - May 2026**

- Planned UAV multispectral campaigns across three growing seasons, capturing 720 rice research plots at ~10-day intervals under six nitrogen rates and multiple cultivars.
- Processed imagery in Pix4D Mapper into radiometrically calibrated 2.5 cm orthomosaics via ground control points and reflectance-panel calibration; derived 19 vegetation indices for yield modeling.
- Benchmarked 12 machine-learning models in Python (R2 = 0.89 validation; nRMSE below 9% on an independent season); first-author journal article plus a co-authored Q1 article on UAS acquisition timing for wheat yield prediction.

**Graduate Teaching Assistant - Louisiana State University | Jan 2024 - May 2024:** Instructed 50+ undergraduates across weekly BIOL 1208 laboratories.

## SELECTED PROJECTS

**UAV Multispectral Imagery Modeling with GIS** - 12-model benchmark (R2 = 0.89) for rice yield; first-author paper in Remote Sensing.

**Gardere Community Hub App** - Experience Builder + Survey123 app; 4 layers, 60+ community resources.

**GIS-Based Fertility Mapping** - IDW fertility surfaces for Nepal cardamom region; peer-reviewed article.

**Parish-Level Offline Field System in GIS** - 16 offline map areas for parish utility disaster response.

**Drone Intelligence for Prediction Modeling with GIS** - UAS flight timing + explainable AI for wheat yield.

**GIS Maturity Assessment and Geospatial Excellence** - Five-pillar assessment + web GIS build for Baton Rouge Green.

## EDUCATION

**M.S., Plant, Environmental and Soil Sciences - Louisiana State University** | May 2026

Focus: GIS & Remote Sensing; Minor in Applied Statistics. Thesis: UAV-based remote sensing and machine learning.

**B.S., Agriculture - Tribhuvan University** | 2021

## SELECTED PUBLICATIONS

Pokharel, R., et al. (2026). Rice Yield Estimation Based on Machine Learning Applied to UAV Remote Sensing Data. Remote Sensing, 18(10), 1575.
Baker, S., et al., incl. Pokharel, R. (2026). Strategically Timed UAS Data Acquisition for Wheat Yield Prediction Using Explainable AI. Smart Agricultural Technology, 15, 102606.
Pokharel, R., et al. (2023). Soil Analysis GIS-Based Fertility Assessment and Soil Mapping of Cardamom Growing Area, Pandam, Ilam, Nepal. IJRASET.

## CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

**Esri MOOCs:** Make an Impact with Enterprise GIS; The ArcGIS Imagery MOOC. **Esri Training:** Experience Builder, ArcGIS API for Python, Drone2Map, ArcGIS Online, Survey123, Imagery & Remote Sensing. **AI:** Claude 101, Claude Code 101, AI Fluency: Framework & Foundations. **In progress:** GISP-E (PreGISP) examination, Nov 22, 2026. **Awards:** Third Place, Research Poster - AI in Agriculture and Natural Resources Conference, Mississippi State University, 2025; Oral Presentation - ASA-CSSA-SSSA Annual Meeting, Salt Lake City, 2025.